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Native AI-based hybrid deep learning for wireless link quality prediction in NTN waterside scenarios

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ABSTRACT

Predicting link quality before establishing communication between transmitter and receiver enhances channel selection. With the advancements in artificial intelligence, prediction is now possible for complex environments such as riverside, maritime and polar regions. This paper evaluates Wi-Fi and LoRa radios, utilizing Received Signal Strength Indicator (RSSI) to understand link quality in riverside environments. The proposed approach compares traditional regression techniques with advanced deep learning models. Error metrics such as Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and Mean Absolute Error (MAE), assess performance. The results demonstrate that ST-LSTM-CNN consistently surpasses other models for Native AI for Non-Terrestrial Networks (NTN).

1. Introduction

The necessity for reliable and intelligent wireless communication systems in riverside environments has been highlighted in recent years by our growing reliance on international trade as well as the rising demands of environmental monitoring and disaster response [1]. Over the past few decades, both maritime and inland waterway transport have experienced significant transformations, driven by advancements in communication technologies [2]. Rivers and lakes are examples of inland waterways that specifically support food and energy security [3]. Modern maritime communication systems integrate satellite, ship-to-submarine, and underwater sensor networks, significantly improving connectivity and overall efficiency [4]. Similarly, inland waterways have seen substantial improvements in wireless technologies, although they present unique challenges that are distinct from terrestrial and open maritime environments [5]. Applications like environmental monitoring, port-to-vessel cooperation, and search and rescue depend on wireless communication [6]. Any failure in communication dependability in such scenarios could have adverse outcomes. River communication systems have incorporated radio frequency (RF) technologies, specifically Long Range (LoRa) and Wireless Fidelity (Wi-Fi), due to their distinct advantages [7]. Wi-Fi networks offer high-speed connectivity at relatively low cost but have limited range capabilities, making them suitable primarily for ship-to-shore or close-range applications [8,9]. Alternatively, LoRa technology supports extensive coverage suitable for long-range communication in complex environments [10]. However, additional regulatory factors such as safety standards, spectrum allocation, and environmental impacts, further complicate the design of robust wireless systems in these unique settings.

Wireless communication in riverside areas encounters specific obstacles due to unpredictable environmental conditions [11]. The presence of water bodies, mobile nodes (such as vessels), and varying topographical features leads to severe signal reflections, refractions, and multipath fading. These conditions significantly deviate from conventional terrestrial communication models, rendering traditional statistical methods inadequate [12,13]. Recognizing these limitations, riverside environments can be categorized under Non-Terrestrial

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Networks (NTNs), as they often lack terrestrial infrastructure and require innovative and intelligent network solutions [14]. Under such constraints, riverside communication scenarios are advancing along with NTN framework that introduces communication systems in environments without permanentterrestrial infrastructure [15]. Particularly, 3GPP specification TS 22.119 and TR 22.819 lists maritime and similar deployments as key NTN cases [16,17].

Though NTN is commonly referred to satellite communications, new applications have been involved in high altitude platforms and edge-assisted systems in geographical locations that are beyond terrestrial connectivity [15]. In this case, riverside environments are not used as a definition of NTN, but as a suitable scenario where key features of NTNs, e.g. link variability and infrastructure independence, may be tested with empirical RF data.

Future 6G networks of communication are anticipated to provide adaptive connectivity in remote or sparse infrastructure regions, e.g., inland deltas, Arctic passages, and river corridors [18]. Riverside settings typically provide representative environments used in these deployments, where signals experience varying conditions due to mobile sources of obstruction as well as variable channel states.

There is also the fact that energy constraints are common in NTN compatible deployments. The 3GPP TR 38.811 covers the architecture of the sustainable, where energy-harvesting modules could be added to enable autonomous node operation where the environment does not provide reliable power distribution [19]. The current work does not project long-duration sensing but shows short-term, low-duty-cycle transmissions as dictated by low-power system design. The deployment vision is shown in Fig. 1 within the 3GPP NTN sustainability roadmap.

This research proposes the ST-LSTM-CNN hybrid deep learning model, specifically designed around native AI principles such as autonomous learning, dynamic adaptability, and robust performance in infrastructure-limited scenarios. The experimental setup incorporates Wi-Fi and LoRa radios in riverside environments consistent with NTN ideal, addressing the knowledge gap concerning wireless link quality prediction under complex conditions and limited infrastructure availability. Through comprehensive experiments, the proposed model demonstrates superior accuracy and adaptability. By aligning closely with native AI concepts, this approach not only tackles immediate challenges but also provides a foundation for future integration into sophisticated, autonomous communication systems anticipated with the advancement of 6G technologies. This work significantly advances intelligent wireless communication capabilities in real-world NTN scenarios. As shown in Fig. 2, the experimental setup for dataset collection involves two communication radios deployed in a riverside environment that aligns with paradigms of NTNs.

2. Related Works

Wireless communication technologies have made significant improvements over the years, particularly in terrestrial and maritime environments, where they play a crucial role in various applications [20]. They emphasize the need for reliable communication systems in smart cities and vehicular networks, while also addressing the critical operations supported by maritime communications, such as navigation and vessel-to-vessel communication [21]. Although the focus is still on terrestrial communication networks, some key players in research and development are directing attention toward other environments, which have completely different wireless communication characteristics

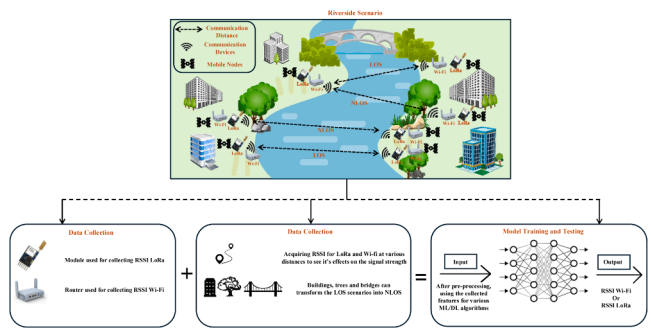

Fig. 2. Wireless Communication Experiment Setup for Riverside Link Quality Prediction.

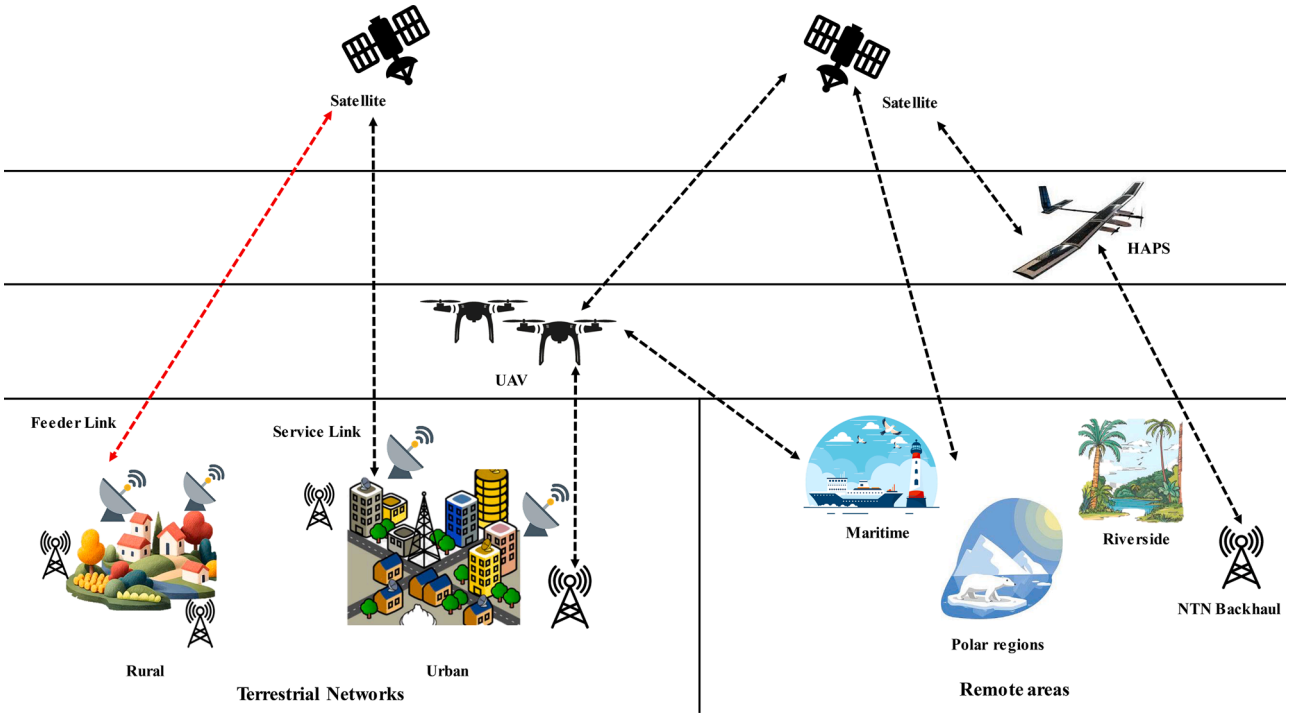

Fig. 1. Deployment vision of 3GPP NTN scenario.

compared to land-based environments [22].

River-side environments represent a unique ecosystem that develops along riverbanks. These habitats exhibit characteristics that blend elements of both terrestrial and maritime ecosystems. A significant factor influencing the ecological dynamics and physical characteristics of these riverbank areas is the presence of water [23]. The reflective surface of the river causes signals to bounce back and forth, resulting in multipath propagation, which combines in the receiver and causes variations in signal strength. Trees, shrubs, and other vegetation near the banks of the river absorb and scatter signals reducing the strength and reliability of wireless links [24]. Bridges, dams, and buildings near the river further complicate the environment by blocking communication paths and introducing obstructed signal conditions, thus deteriorating the quality of signals even further [25,26]. These factors result in high signal variability, multipath propagation, and frequent transitions between line-of-sight (LOS) and non-line-of-sight (NLOS) conditions, factors that demand adaptive and context-aware communication strategies [27].

However, despite the distinct challenges posed by these environments, prior studies have mostly adopted traditional statistical methods or rule-based systems, which fall short in capturing signal instability under highly dynamic conditions. These approaches often rely on signal averaging, which obscures important variance-related features. In contrast, AI-based approaches can incorporate statistical features such as standard deviation and coefficient of variation (CV), enabling context-aware modeling of unstable RF environments [28].

Table 1 is a comparison of other (non-AI, legacy) and AI-based approaches to wireless link analysis in terms of the methodological differences, intended environment, and performance metrics. Non-AI techniques tend to be either threshold-based or based on fixed heuristics, neither of which are flexibility required in time-varying, multi-path dominated situations such as riversides. Methods based on AI, and especially deep learning models, can learn the temporal and spatial variation of signals and can achieve better generalisation under different circumstances. A summary of relevant prior research is presented in Table 2, detailing the methodologies, key findings, and limitations of each study. This table gives a comparative study which is useful in defining the research gap.

Deep learning, while having proved itself useful for handling many other complex communication problems, is still very much unexplored for communications at the riverside. So far, existing studies on link quality prediction have mainly tried to use terrestrial environments to address spatial data or traffic patterns [29,30]. Most of the maritime research has been on optimal satellite and long-distance communication, leaving out small-scale and dynamic scenarios such as those presented by riversides [31]. These studies provide valuable insights into how deep learning models can be adapted for riverside environments, particularly in applications such as disaster management, smart transportation, and environmental monitoring. One of the most critical challenges in riverside environments is disaster response, particularly in mitigating the impact of floods. Reliable communication networks and AI-driven analysis play a crucial role in early warning systems, enabling real-time monitoring and decision-making [32].

Table 1
Comparison of legacy vs. AI-based approaches for wireless link analysis.

Feature/Aspect	Legacy Methods (Non-AI) [A]	AI-Based Approaches [B]
Methodology	Rule-based, thresholding, heuristics [37]	Supervised learning, deep learning, hybrid models [38]
Signal Variance Handling	Often ignored or averaged out [39]	Explicitly used (e.g., std, CV as features) [40]
Environmental Awareness	Static; limited to pre-defined conditions [41]	Dynamic; learns from environment-specific patterns [42]
Adaptability	Low (hard-coded rules) [37]	High (trained to generalize over variability) [38]
LOS/NLOS Consideration	Rarely formalized [43]	Incorporated as features for prediction[40]

Flood events have become more frequent and severe, primarily driven by changing climatic patterns, posing risks to lives, farmlands, and infrastructure [33]. Efficient remote sensing systems and IoT-based water level sensors deployed along riverbanks are crucial for capturing real-time and historical data to enhance flood prediction accuracy [34]. Although terrestrial and maritime wireless technologies have advanced, riverside environments present unique NTN challenges that remain underexplored. Preliminary efforts often rely on traditional statistical methods rather than AI-driven approaches [35,36]. To address this gap, this study proposes a native AI-based deep learning framework for resilient riverside NTN communication networks.

3. Methodology

Wi-Fi and LoRa radios are considered for the proposed wireless link quality prediction. Experiments are conducted in a real-time riverside scenario, with transmitters and receivers separated by varying distances and conditions. While LoRa operated in the KR920–923 band, Wi-Fi operated in the 2.4 GHz band.

The communication modules used in this study were not manually adjusted in terms of transmission power. All hardware operated under default configurations, compliant with IEEE and manufacturer standards. The resulting data reflect realistic transmission conditions, without artificial constraints on output power. This clarification distinguishes the study from prior models that assume fixed-power LoRa links. To achieve the objective, RSSI values are gathered at various distances in both LOS and NLOS scenarios . Various experiments were carried out over two days. Obtaining RSSI values from Wi-Fi and LoRa radios in maritime settings is part of the data collection process. In this study, LOS is defined as a direct visual path between transmitter and receiver, while NLOS refers to any obstructed scenario due to terrain or infrastructure. This classification is applied consistently across experiments and serves as a contextual feature in model training. Variations in LOS/NLOS states correlate with fluctuations in RSSI, anomaly occurrences, and link instability. Post collection, to deal with missing values and outliers, the pre-processed data is used as input for deep learning models. The dataset, comprising 5458 samples and 7 attributes, is divided into 80% training and 20% testing subsets. Few samples of the dataset are shown in the Table 4. This experimental design facilitates the application of Native AI, wherein models are trained on environment-specific data to inherently capture the spatiotemporal characteristics of riverside conditions. Such an approach enhances the model's ability to generalize within NTN scenarios by utilizing localized data patterns for robust and adaptive link quality prediction. Table 3 lists the features, and Fig. 3 illustrates the experiment process flow.

To achieve the intended outcomes of this research, a variety of machine learning and deep learning techniques are employed. To evaluate model performance, widely used statistical error metrics such as RMSE, MSE, and MAE are applied, ensuring a comprehensive assessment of predictive accuracy [44–48]. The upcoming subsections provide detailed descriptions of these models.

Before model development, an exploratory data analysis was conducted to understand feature interactions. Correlation analysis and Recursive Feature Elimination highlighted that distance and environment are important predictive factors for Wi-Fi and LoRa. These findings contribute significantly to developing a hybrid model which can adapt to dynamic NTN conditions [49].

Our initial investigation utilized a standalone ST-LSTM-CNN architecture, which demonstrated strong potential for link quality prediction by capturing spatiotemporal characteristics in the data. To further enhance the model's predictive power and rigorously test its generalizability, we developed an advanced hybrid stacking framework.

This hybrid approach integrates the ST-LSTM-CNN as a meta-learner that operates on an enriched feature set. The original features (Distance, Environment) are augmented with k-fold predictions generated by a

Table 2
Summary of key related works.

Ref.	Focused Area	Methodology / Algorithm	Environment / Dataset	Key Findings	Limitations / Gaps	Relevance to This Study/ Contribution
[50]	Link Quality (LQ) prediction	Long Short-Term Memory (LSTM-RNN), Gated Recurrent Unit (GRU)	The data was collected from the Funkfeuer Wien WCN in Vienna, Austria.	High Accuracy; Superiority over Traditional Methods	The authors suggest that performance should be assessed on additional real-world datasets in the future	Demonstrates Machine Learning(ML) feasibility for wireless link quality prediction
[30]	AI-based channel selection for wireless networks	KNN, Random Forest, SVM	Terrestrial Wi-Fi and LoRa testbeds	KNN achieved 0.98 precision in dynamic channel selection	Generalizability to complex NTN untested	Shows AI potential in handling interference and channel variability
[54]	The study focuses on ship trajectory prediction using Automatic Identification System (AIS) data.	GRU, Bi-directional GRU (Bi-GRU), and Transformer	The study uses AIS data from a normal channel (Chengshan Jiao Promontory), an area with complex traffic (Zhoushan Archipelago), and a busy port area (Caofeidian port).	In general, deep learning methods showed higher prediction accuracy than machine learning methods, especially as the complexity of the ship trajectories increased.	The paper acknowledges that future work should focus on improving accuracy by integrating multi-source information (like weather data), which this study did not do. The authors state they do not have permission to share the raw data used in the study.	Systematically test and compare twelve advanced prediction methods side-by-side using multiple real-world datasets and a robust set of evaluation metrics.
[55]	Focuses on indoor object localization by enhancing the use of the Received Signal Strength Indicator (RSSI) from Wi-Fi signals.	KNN, Random forest, SVM and Decision Trees.	The data was collected in two adjacent rooms within an academic building. Three Wi-Fi access points were placed in one of the rooms to generate the RSSI signals	The machine learning models, particularly tree-based methods, were highly effective .	The study used a small number of devices and access points.	The study validates the use of advanced feature engineering techniques like Dynamic Time Warping (DTW).
[38]	The paper focuses on analyzing the ML process for Wireless Link Quality Estimation (LQE).	Logistic Regression, Decision Trees, Random Forest, SVM, and Multi-layer Perceptron.	The study uses the Rutgers trace-set, which was identified as the most suitable publicly available dataset for LQE research.	The study's main conclusion is that the initial stages of the ML pipeline are more critical than the final choice of algorithm.	The paper highlights that the broader LQE research field lacks standardized datasets	This paper provides critical insights for researchers by demonstrating where to focus efforts when building an ML-based LQE model.
[56]	The main goal is to detail a systematic process for transforming unstructured RSSI signal data into a clean, structured dataset suitable for predictive modeling tasks.	The study proposes a two-stage data preparation architecture consisting of data construction and data cleaning.	The RSSI data was collected from the Dinamika Bangsa University Campus building in Indonesia, specifically across the 1st and 4th floors.	The systematic data preparation process successfully produced a high-quality dataset ready for machine learning applications.	This paper is a methodological report focused exclusively on the data preparation process and does not involve building or testing a predictive model.	This paper's primary contribution is providing a clear, step-by-step blueprint for preparing raw RSSI data, which is a critical but often overlooked stage in machine learning projects.

Table 3
Dataset description.

Features	Description
Distance	12.635~82.4145 (metres)
Environment	LOS/NLOS
RSSI Wi-Fi	-178.5~-63.5 (dBm)
RSSI LoRa	-105.9~-57.7 (dBm)

Table 4
Sample data from riverside experiment.

S. No.	Date	Timestamp	Environment	Distance	RSSI Wi-Fi	RSSI LoRa
1.	05-10-2023	12:32:06	NLOS	78.49	-170	-89
2.	05-10-2023	11:28:22	LOS	30.78	-85.05	-78.75
3.	08-10-2023	16:00:20	LOS	15.225	-71.45	-95.65
4.	05-10-2023	12:00:24	LOS	12.635	-73.2	-65.55
5.	08-10-2023	16:12:00	NLOS	21	-84	-95

diverse set of powerful base models, namely Random Forest and Gradient Boosting [57,58]. By training the ST-LSTM-CNN on the combined insights from these different models, the framework is designed to capture more complex patterns and produce a more robust final prediction. The performance of this hybrid model was evaluated using a strict k-fold cross-validation to ensure an unbiased assessment and prevent any potential data leakage, thereby providing a true measure of its generalizability on unseen data.

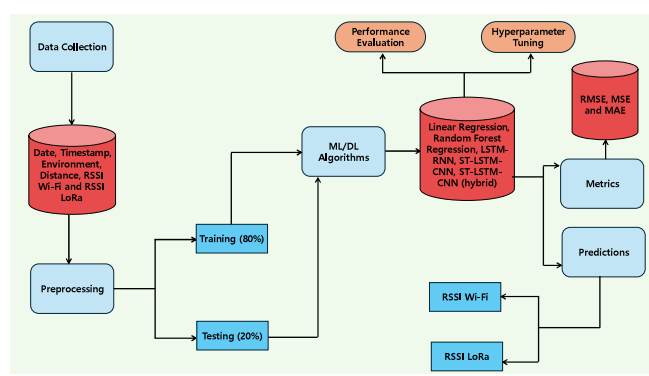

Fig. 3. Experimental Workflow for RSSI Prediction.

3.1. Recurrent Neural Networks (RNNs)

RNNs are designed for recognizing patterns in sequential data. Unlike traditional feedforward networks, RNNs incorporate cyclic

connections, enabling them to retain information from prior input and maintain temporal dependencies. This ability to preserve historical context makes RNNs highly effective for applications that require an understanding of sequential dependencies, such as time-series forecasting, speech processing, and text-based predictions.

3.2. Linear Regression

A linear regression model stands as the starting point for Wi-Fi and LoRa RSSI prediction because it analyzes distance variations alongside environmental factors. Linear regression functions as a basic yet successful model to identify linear patterns which connect the predictor variables to the target values.

3.3. Polynomial Regression

The polynomial regression model extends linear regression through non-linear feature interactions to detect advanced data relationships in datasets. The model generates degree 2 polynomial features to add interaction and squared terms between distance and environment and RSSI variations to capture non-linear behaviours. After transformation, the features enter a linear regression model which produces predictions based on fitted polynomial features.

3.4. Random Forest Regression

By leveraging ensemble learning, Random Forest enhances decision tree-based predictions, minimizing overfitting while optimizing generalization. The algorithm builds multiple decision trees trained on different data subsets, and the final output is obtained by averaging their predictions. The procedure facilitates the detection of intricate relationships in wireless systems since it handles signal strength fluctuations caused by multipath propagation combined with environmental interference and changing environmental factors.

3.5. LSTM-RNN

The model leverages the power of LSTM networks, which are well suited to learning temporal dependencies in sequence data. The input layer receives the normalized set of features of distance and environmental conditions. The input features and target variables are normalized to a range of 0 and 1 using MinMaxScaler so that the model learns the patterns efficiently without being affected by varying feature scales. An initial LSTM layer consisting of 50 units processes sequential input, effectively modeling long-term signal dependencies. To meet LSTM input format requirements, data is structured into a three-dimensional format, comprising samples, time steps, and features. This enables the model to identify temporal variations in signal strength in relation to distance and environmental conditions, particularly in riverside communication scenarios. A fully connected Dense layer with two neurons is implemented to generate RSSI predictions for Wi-Fi and LoRa. Since this is a regression-based task, the output layer utilises a linear activation function to ensure the generation of continuous values.

3.6. ST-LSTM-CNN

ST-LSTM-CNN, which is proposed, is a unique architecture that captures and integrates spatial and temporal characteristics from sequential data, represented in Fig. 4. The model accepts sequential data as input, which is, distance, environment, RSSI Wi-Fi and RSSI LoRa. The initial Conv1D layer with 128 filters and a kernel size of 1 is applied to the input sequence. This layer's purpose is to identify local patterns within the sequence by extracting high-level spatial information from the data. Following the Conv1D layer, a MaxPooling1D layer with a pool size of 1 is applied to down-sample the feature maps. The output from the MaxPooling1D layer is reshaped to make it compatible with the

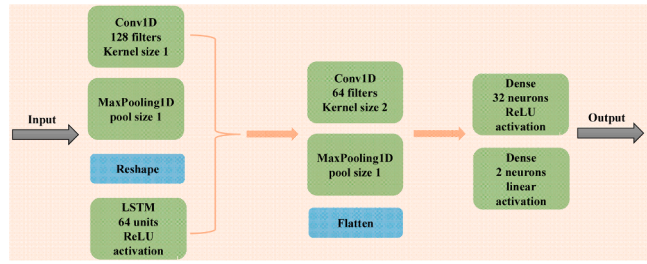

Fig. 4. Structure of the Proposed ST-LSTM-CNN Model.

LSTM layer. The reshaped data is fed into an LSTM layer with 64 units. Because of its built-in memory cells and gating mechanisms, long-range relationships in sequential data are easily captured by LSTMs. ReLU activation is applied to enhance the model's capability of capturing intricate temporal dependencies. The architecture incorporates parallel convolutional pathways, further refining feature extraction. A fully connected dense layer comprising 32 neurons utilizes ReLU activation to model complex feature interactions. Finally, the output layer consists of two neurons with a linear activation function, ensuring the generation of continuous values, which is ideal for regression-based tasks.

3.7. ST-LSTM-CNN(Hybrid)

The Hybrid ST-LSTM-CNN is an advanced architecture designed to enhance predictive accuracy by leveraging a stacking ensemble methodology represented in Fig. 5. The hybrid model, unlike the baseline one, takes an augmented feature set as input. This ensemble consists of original predictive factors (Distance and Environment) supplemented with the k-fold prediction of two strong base learners: Random Forest and Gradient Boosting. This approach provides the core deep learning model with a richer, more diverse set of information, allowing it to learn from the pre-processed insights of established ensemble techniques. The architecture begins by processing this hybrid input through two parallel pathways to capture both spatial and temporal characteristics: A spatial path utilizes a Conv1D layer (kernel size of 1) with 512 filters that extract local patterns out of the feature set and a MaxPooling1D layer that down-samples. To model the sequential dependencies and long-range relationships in the data using a temporal path, an LSTM layer with 256 units is used.

The resulting features of the two paths will be concatenated and fed into another Conv1D layer of 256 filters to sharpen the acquired features. The data is then flattened and fed through a stack of Dense layers with ReLU activation which fully connect and follow the pattern (128 and 64 neurons respectively) to represent intricate interactions. The output layer is two neurons with a linear activation unit to obtain continuous RSSI predictions of Wi-Fi and LoRa.

4. Results

Given the absence of fixed infrastructure and the dynamically changing conditions in riverside areas, the experimental setup

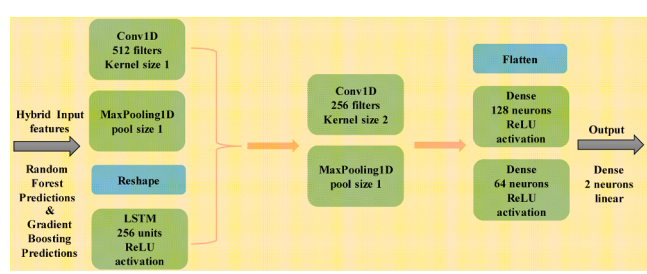

Fig. 5. Structure of the Proposed ST-LSTM-CNN(Hybrid) Model.

effectively represents a real-world NTN scenario. This context allows for a realistic evaluation of model performance under conditions where conventional methods typically underperform. In evaluating the predictive capabilities of various machine learning models for Wi-Fi and LoRa, performance is measured using RMSE, MSE, and MAE. As summarized in Table 5, the results highlight that the proposed ST-LSTM-CNN model surpasses other models, delivering the lowest error values across all three metrics.

The ST-LSTM-CNN model's enhanced performance stems from its ability to model both spatial and temporal dependencies within the dataset. Specifically, the CNN segment captures local spatial patterns, while the LSTM component models long-term temporal relationships. By integrating these components, the model achieves greater prediction accuracy.

The 3D plots in Figs. 6–8 visually represent the performance of different models in terms of three evaluation metrics: RMSE, MSE, and MAE. Traditional machine learning models show significantly higher values, demonstrating their limitations in handling sequential dependencies in the data. The RNN and LSTM-RNN models show improved performance over traditional models, suggesting that incorporating temporal dependencies improves accuracy. The analysis of these figures validates that the ST-LSTM-CNN model is the most effective in predicting between Wi-Fi and LoRa networks. It demonstrates lower values compared to traditional and other models.

The performance of several models was evaluated, with the Random Forest serving as a strong benchmark due to its proficiency with tabular data. Our baseline ST-LSTM-CNN model also demonstrated highly effective performance in initial tests.

The proposed hybrid ST-LSTM-CNN, evaluated under a rigorous k-fold cross-validation framework, achieved an average RMSE of 5.6742. For comparison, the highly optimized Random Forest model achieved an RMSE of 5.6094. Table 6 describes the comparative performance of Link Quality Predictions Models.

The key finding is the convergence in performance between these two distinct architectures. The negligible difference in their error metrics indicates that the hybrid ST-LSTM-CNN can perform on par with one of the most powerful traditional models for this task. This result is significant, as it validates the generalizability and robustness of our proposed deep learning architecture. It successfully captures the complex, non-linear relationships within the data to a degree that is competitive with leading ensemble methods, confirming its viability as an advanced solution for link quality prediction in challenging NTN environments.

5. Conclusions

This research demonstrates the potential of advanced machine learning models, particularly ST-LSTM-CNN, for accurate link quality prediction in challenging maritime environments. By leveraging Wi-Fi and LoRa radios and incorporating the RSSI parameter, we have achieved significant improvements over existing approaches, including traditional statistical methods and other AI-based models. The superior performance of ST-LSTM-CNN, as evidenced by the lowest error rates across RMSE, MSE, and MAE, highlights its ability to effectively capture the complex spatiotemporal dependencies inherent in riverside environments.

Table 5
Comparison of different models with evaluation metrics.

Model	RMSE	MSE	MAE
RNN [50]	0.1189	0.0146	0.0863
Linear Regression [51]	0.3488	0.1216	0.3898
Polynomial Regression [52]	0.2323	0.0540	0.2811
Random Forest Regression [53]	0.1792	0.0321	0.2042
LSTM-RNN [50]	0.1256	0.0157	0.0923
ST-LSTM-CNN [proposed]	0.1038	0.0107	0.0779

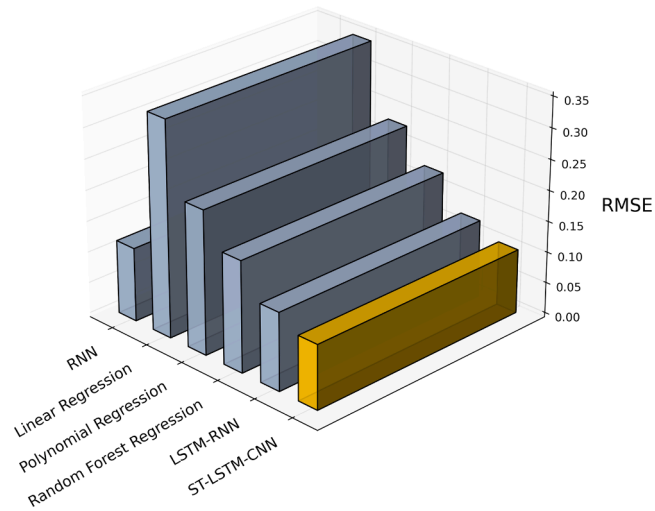

Fig. 6. RMSE Performance.

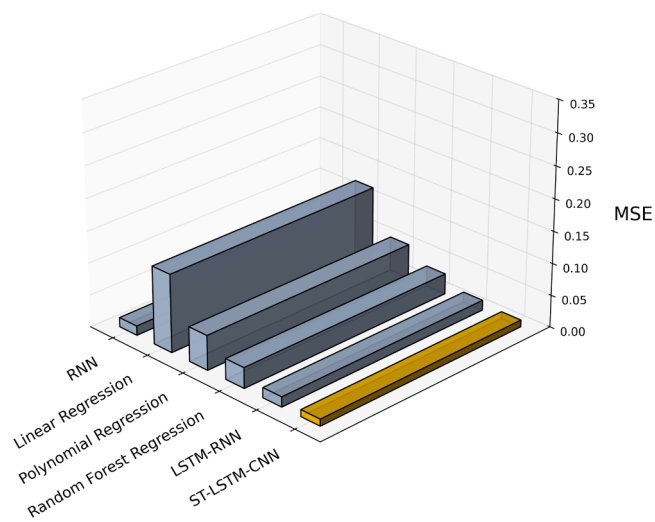

Fig. 7. MSE Performance.

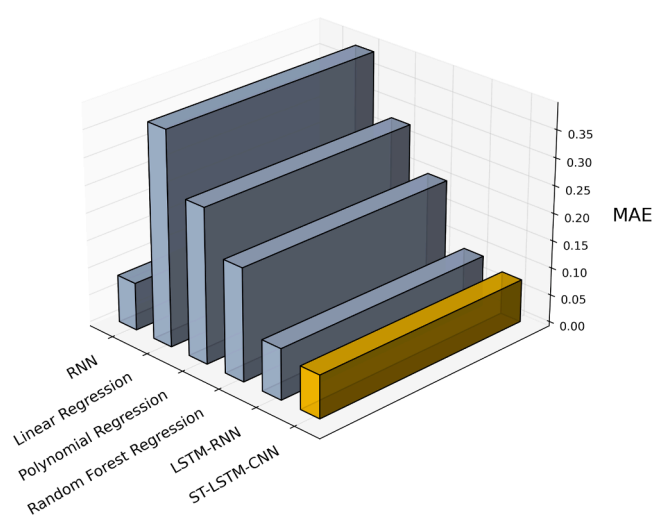

Fig. 8. MAE Performance.

Table 6
Comparative performance of link quality prediction models.

Model	RMSE (Mean ± Std)	MSE (Mean ± Std)	MAE (Mean ± Std)
Linear Regression	10.5301 ± 0.1469	110.9042 ± 3.1046	8.7145 ± 0.1414
Polynomial Regression (Deg=2)	7.4501 ± 0.1409	55.5238 ± 2.0902	6.0744 ± 0.0856
LSTM-RNN	8.0298 ± 0.1387	64.4973 ± 2.2320	6.1852 ± 0.1410
Random Forest Regression	5.6094 ± 0.0325	31.4660 ± 0.3635	4.7605 ± 0.0337
ST-LSTM-CNN [Hybrid]	5.6742 ± 0.0325	32.1998 ± 0.3687	4.7991 ± 0.0199

Through a strict k-fold cross-validation process, designed to ensure scientific rigor and prevent data leakage, we found that the error metrics of both models were closely aligned. This result is significant for two primary reasons:

It validates the viability of our deep learning approach. The ST-LSTM-CNN's ability to match the performance of a leading tree-based model confirms its capacity to effectively capture the complex spatio-temporal dependencies inherent in wireless signal propagation in these unique environments.

It emphasizes the generalizability of the model. Achieving this level of performance under rigorous cross-validation demonstrates that the model is not merely memorizing the training data but is learning the underlying patterns in a way that translates to unseen data.

This work establishes the hybrid ST-LSTM-CNN as a powerful and validated solution for this prediction task. The true advantage of this deep learning framework lies in its future potential.

The results represent an important advancement toward achieving intelligent wireless communication systems capable of reliable performance in non-terrestrial network (NTN) environments. The framework lays a promising foundation for future integration of native AI approaches into communication infrastructure. Future research directions include extending the model to extreme environments such as polar regions and leveraging emerging deep-learning methodologies to enhance feature extraction and performance in real-time native AI-enabled communication contexts.

CRediT authorship contribution statement

Shrutika Sinha: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **G. Pradeep Reddy:** Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sea-Moon Kim:** Writing – review & editing. **Soo-Hyun Park:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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